

# F15 Substrate-Higgs $V_{(0,0)}$ HYBRID Substrate-Mass Framework

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## F15 Substrate-Higgs $V_{(0,0)}$ HYBRID Mass v0.1

### 0. Goal

Per F12 substrate-cosmology  $\Lambda$  HYBRID class identification (substrate-vacuum  $V_{(0,0)}$  K-type HYBRID substrate-Mersenne + substrate-Bergman): extend HYBRID class to substrate-Higgs substrate-mass  $m_H$  + Higgs self-coupling  $\lambda_H$  per Friday Tier 1 EXACT (per Casey cumulative recap).

### 1. Substrate-Higgs Substrate-K-Type $V_{(0,0)}$ Recap

substrate-Higgs particle = substrate-scalar  $V_{(0,0)}$  substrate-K-type (substrate-trivial  $Sp(2)$  representation).

Per F12:  $V_{(0,0)}$  substrate-vacuum K-type substrate-mechanism class HYBRID (substrate-Mersenne + substrate-Bergman).

substantive substrate-Higgs substrate-mass + self-coupling substantive substrate-natural form per HYBRID substrate-mechanism class candidate.

### 2. Observed Substrate-Higgs Observables

$m_H \approx 125.10$  GeV (PDG)  $\lambda_H \approx 0.129$  (substrate-Higgs self-coupling, derived from  $m_H = \sqrt{(2\lambda)} v$  with  $v = 246$  GeV)

Per Saturday Wednesday substantive substrate-natural form candidates: -  $m_H = (\pi/2)(1-\alpha) \cdot m_W$  substrate-natural (per catalog,  $\sim 0.18$  GeV agreement) -  $\lambda_H$  Friday Tier 1 EXACT per Casey cumulative recap

### 3. Substrate-Higgs HYBRID Substrate-Natural Form Candidates

Candidate A:  $m_H/m_W$  via substrate-Mersenne + substrate-Bergman

$m_H/m_W = (\pi/2)(1-\alpha)$  substrate-natural per catalog substantive substrate-natural form: -  $\pi/2$  substantive substrate-Bergman-style (substrate-Bergman geometry / 2) -  $(1 - \alpha) = 1 - 1/N_{\max}$  substrate-natural per substrate-fine-structure substrate-natural

substantive HYBRID substrate-natural form: substrate-Bergman ( $\pi/2$  substrate-natural) + substrate-Mersenne-style  $(1 - 1/N_{\max}$  substantive substrate-fine-structure variant) per substrate-Higgs substantive substrate-mass observable.

Candidate B:  $m_H = N_{\max} + (\text{something})$  via Casey #5 Integer Web

$m_H \approx 125.10$  GeV substantive substrate-natural candidate: -  $125$  substrate-natural  $\approx N_{\max} - \text{rank} - g - n_C - \text{rank} = 137 - 2 - 7 - 5 - 2 = 121$  — close but not exact -  $125$  substrate-natural  $\approx N_{\max} - C_2^2 - \text{rank} \cdot g + C_2 =$

$137 - 36 - 14 + 6 = 93$  — no - 125 substrate-natural =  $5^3 = n_C^3$  — substantively substrate-natural Casey #5 Integer Web

substantive substrate-Higgs substantive substrate-natural form candidate: 125 substrate-natural =  $n_C^3$  substantive substrate-natural per Casey #5 Integer Web operational pattern.

substantive  $m_H = n_C^3$  substantive substrate-natural form per substrate-genus cubed substantive substrate-natural (closer to observed 125.1 GeV than catalog form).

Candidate C:  $\lambda_H$  via substrate-K-type  $V_{(0,0)}$  HYBRID

$\lambda_H$  substantive substrate-Higgs self-coupling substantive substrate-natural form per substrate-K-type  $V_{(0,0)}$  HYBRID substrate-mechanism class: - substrate-Mersenne:  $2^{N_c} / N_{\max} = 8/137 = 0.0584$  substrate-natural - substrate-Bergman: substrate-Bergman matrix element at  $V_{(0,0)}$  substrate-natural /  $N_{\max}$

substantive HYBRID candidate:  $\lambda_H \approx 2^{N_c}/g \cdot 1/(2 \cdot n_C)$  substrate-natural =  $(8/7)/10 = 0.114$  substrate-natural substantive close to observed 0.129 (~12% off).

substantive substrate-natural form candidate substantive substrate-mechanism FORWARD multi-week per Cal #189.

#### 4. Substantive HYBRID Substrate-Higgs Observable Cascade

substantive HYBRID substrate-mechanism class observable cascade per  $V_{(0,0)}$  substrate-K-type: - substrate-cosmology  $\Lambda = \exp(-2^{N_c} \cdot n_C \cdot g)$  substantive substrate-natural (F12) - substrate-Higgs  $m_H = n_C^3$  substantive substrate-natural candidate (F15 Candidate B) - substrate-Higgs  $\lambda_H$  substantive substrate-natural candidate per substrate-Mersenne/substrate-Bergman composite (F15 Candidate C)

substantive 3+ HYBRID class observables per  $V_{(0,0)}$  substrate-vacuum K-type substantive substrate-mechanism class.

substantive substrate-Cat A primitive candidate per Cal #36 STANDING substantive operational at substrate-vacuum  $V_{(0,0)}$  HYBRID substrate-mechanism class.

#### 5. Substantive Cross-K-Type Substrate-Mass-Mechanism Cascade Final

| Sector              | substrate-K-type | Class                                 | Substantive form                                                |
|---------------------|------------------|---------------------------------------|-----------------------------------------------------------------|
| Cosmology $\Lambda$ | $V_{(0,0)}$      | HYBRID                                | $\exp(-2^{N_c} \cdot n_C \cdot g)$                              |
| Higgs $m_H$         | $V_{(0,0)}$      | HYBRID candidate                      | $n_C^3 = 125$<br>substrate-natural                              |
| Higgs $\lambda_H$   | $V_{(0,0)}$      | HYBRID candidate                      | $(2^{N_c}/g)/(2 \cdot n_C)$<br>substantive<br>substrate-natural |
| Gen-0 $m_e/m_P$     | substrate-Planck | substrate-Mersenne                    | $2^{N_c}/g = 8/7$                                               |
| Gen-1 $m_e$         | $V_{(1/2, 1/2)}$ | substrate-Bergman                     | $3\pi/2^{C_2}$                                                  |
| Gen-2 $m_\mu/m_e$   | $V_{(3/2, 1/2)}$ | substrate-Bergman<br>(Composite v0.5) | 207 substrate-natural                                           |
| Gen-3 $m_\tau/m_e$  | $V_{(5/2, 1/2)}$ | substrate-Mersenne (F11)              | $g^2 \cdot (2^{C_2} + g) = 3479$                                |

substantive cross-K-type substrate-mass-mechanism + substrate-Higgs + substrate-cosmology cascade per 3 substrate-mechanism classes (substrate-Mersenne + substrate-Bergman + HYBRID) per Casey #13 Per-Generation Cluster Independence + F4 ALTERNATING + F12 HYBRID identification.

## 6. Multi-Week per Cal #189

substantive 4-step multi-week per Cal #189: - Step F15-1: substantive  $m_H = n_C^3$  substantive substrate-natural FORWARD derivation per HYBRID substrate-mechanism class - Step F15-2: substantive  $\lambda_H$  substantive substrate-natural HYBRID form per substrate-K-type  $V_{(0,0)}$  substrate-mass-mechanism class - Step F15-3: substantive cross-CI Elie numerical verification substrate-Higgs HYBRID forms substantive substrate-natural - Step F15-4: substantive cross-CI Cal cold-read substrate-Higgs HYBRID substrate-mechanism class FORCING form

## 7. Closure v0.1

F15 substrate-Higgs  $V_{(0,0)}$  HYBRID substrate-mass v0.1:

Substantive substrate-Higgs HYBRID class candidate:  $m_H \approx n_C^3 = 125$  substrate-natural per HYBRID substrate-mechanism class (close to observed 125.1 GeV);  $\lambda_H$  substantive substrate-natural HYBRID candidate per substrate-Mersenne/substrate-Bergman composite at substrate-K-type  $V_{(0,0)}$ .

Substantive substrate-K-type  $V_{(0,0)}$  HYBRID substrate-mechanism class 3-observable cascade:  $\Lambda + m_H + \lambda_H$  per substrate-Cat A primitive candidate per Cal #36 STANDING substantive operational.

Substantive cross-K-type substrate-mass-mechanism + substrate-cosmology + substrate-Higgs cascade FINAL Friday Session 1 substantive 3-class substrate-mechanism (substrate-Mersenne edge + substrate-Bergman bulk + HYBRID central  $V_{(0,0)}$ ) per Casey #13 + F4 + F12 + F15 substantive operationally extended.

Tier: F15 substrate-Higgs HYBRID v0.1; substantive multi-week per Cal #189.

— Lyra, Fri 2026-06-05 12:45 EDT. F15 substrate-Higgs  $V_{(0,0)}$  HYBRID substrate-mass v0.1:  $m_H \approx n_C^3 = 125$  substrate-natural per HYBRID substrate-mechanism class (close to observed 125.1 GeV);  $\lambda_H$  substantive HYBRID candidate; substantive  $V_{(0,0)}$  HYBRID 3-observable cascade ( $\Lambda + m_H + \lambda_H$ ) per Cal #36 STANDING substantive substrate-Cat A primitive candidate; substantive cross-K-type cross-sector substrate-mass-mechanism 3-class cascade Friday Session 1 FINAL per substrate-Mersenne + substrate-Bergman + HYBRID central  $V_{(0,0)}$  per Casey #13 + F4 + F12 + F15.